

Born approximation 1

Derive the Born approximation

$$\psi(r, \theta, \phi) = -\frac{m}{2\pi\hbar^2} \frac{\exp(ikr)}{r} \int \exp\left(\frac{i\mathbf{p} \cdot \mathbf{x}}{\hbar}\right) V(\mathbf{x}) d\mathbf{x}$$

where

$$k = \frac{\sqrt{2mE}}{\hbar}$$

Vector $\mathbf{p}$ is momentum transfer such that

$$\mathbf{p} = \mathbf{p}_i - \mathbf{p}_f$$

and

$$p^2 = |\mathbf{p}|^2 = 4mE(1 - \cos \theta)$$

§1

Start with the Schrodinger equation

$$-\frac{\hbar^2}{2m} \nabla^2 \psi(\mathbf{x}) + V(\mathbf{x})\psi(\mathbf{x}) = E\psi(\mathbf{x})$$

Multiply through by $-2m/\hbar^2$ to obtain

$$\nabla^2 \psi(\mathbf{x}) - \frac{2m}{\hbar^2} V(\mathbf{x})\psi(\mathbf{x}) = -\frac{2mE}{\hbar^2} \psi(\mathbf{x})$$

Interchange E and V terms.

$$\nabla^2 \psi(\mathbf{x}) + \frac{2mE}{\hbar^2} \psi(\mathbf{x}) = \frac{2m}{\hbar^2} V(\mathbf{x})\psi(\mathbf{x})$$

Let

$$k = \frac{\sqrt{2mE}}{\hbar}$$

and rewrite as

$$(\nabla^2 + k^2) \psi(\mathbf{x}) = \frac{2m}{\hbar^2} V(\mathbf{x})\psi(\mathbf{x})$$

By the sifting property of $\delta(\mathbf{x} - \mathbf{y})$ we can make the right-hand side an integral over $\mathbf{y}$.

$$(\nabla^2 + k^2) \psi(\mathbf{x}) = \frac{2m}{\hbar^2} \int \delta(\mathbf{x} - \mathbf{y}) V(\mathbf{y})\psi(\mathbf{y}) d\mathbf{y}$$

Let $G(\mathbf{x})$ be a Green's function such that

$$(\nabla^2 + k^2) G(\mathbf{x}) = \delta(\mathbf{x})$$

Then by substitution

$$(\nabla^2 + k^2) \psi(\mathbf{x}) = \frac{2m}{\hbar^2} \int (\nabla^2 + k^2) G(\mathbf{x} - \mathbf{y}) V(\mathbf{y}) \psi(\mathbf{y}) d\mathbf{y}$$

By linearity the integral can be factored as

$$(\nabla^2 + k^2) \psi(\mathbf{x}) = (\nabla^2 + k^2) \frac{2m}{\hbar^2} \int G(\mathbf{x} - \mathbf{y}) V(\mathbf{y}) \psi(\mathbf{y}) d\mathbf{y}$$

Hence

$$\psi(\mathbf{x}) = \frac{2m}{\hbar^2} \int G(\mathbf{x} - \mathbf{y}) V(\mathbf{y}) \psi(\mathbf{y}) d\mathbf{y}$$

The solution for $G(\mathbf{x})$ is

$$G(\mathbf{x}) = -\frac{\exp(ik|\mathbf{x}|)}{4\pi|\mathbf{x}|}$$

Hence

$$\psi(\mathbf{x}) = -\frac{m}{2\pi\hbar^2} \int \frac{\exp(ik|\mathbf{x} - \mathbf{y}|)}{|\mathbf{x} - \mathbf{y}|} V(\mathbf{y}) \psi(\mathbf{y}) d\mathbf{y}$$

For scattering experiments we can use the approximation

$$\frac{\exp(ik|\mathbf{x} - \mathbf{y}|)}{|\mathbf{x} - \mathbf{y}|} V(\mathbf{y}) \approx \frac{1}{|\mathbf{x}|} \exp\left(ik|\mathbf{x}| - \frac{ik\mathbf{x} \cdot \mathbf{y}}{|\mathbf{x}|}\right) V(\mathbf{y})$$

Hence

$$\psi(\mathbf{x}) = -\frac{m}{2\pi\hbar^2} \frac{\exp(ik|\mathbf{x}|)}{|\mathbf{x}|} \int \exp\left(-\frac{ik\mathbf{x} \cdot \mathbf{y}}{|\mathbf{x}|}\right) V(\mathbf{y}) \psi(\mathbf{y}) d\mathbf{y}$$

For a first-order approximation we substitute a plane wave for $\psi(\mathbf{y})$.

$$\psi(\mathbf{y}) = \exp\left(\frac{ik\mathbf{x}' \cdot \mathbf{y}}{|\mathbf{x}'|}\right)$$

Hence

$$\psi(\mathbf{x}) = -\frac{m}{2\pi\hbar^2} \frac{\exp(ik|\mathbf{x}|)}{|\mathbf{x}|} \int \exp\left(\frac{ik\mathbf{x}' \cdot \mathbf{y}}{|\mathbf{x}'|} - \frac{ik\mathbf{x} \cdot \mathbf{y}}{|\mathbf{x}|}\right) V(\mathbf{y}) d\mathbf{y}$$

Let $\mathbf{p}$ be momentum transfer

$$\mathbf{p} = \mathbf{p}_i - \mathbf{p}_f = \hbar k \left(\frac{\mathbf{x}'}{|\mathbf{x}'|} - \frac{\mathbf{x}}{|\mathbf{x}|} \right)$$

Then by substitution

$$\psi(\mathbf{x}) = -\frac{m}{2\pi\hbar^2} \frac{\exp(ik|\mathbf{x}|)}{|\mathbf{x}|} \int \exp\left(\frac{i\mathbf{p} \cdot \mathbf{y}}{\hbar}\right) V(\mathbf{y}) d\mathbf{y}$$

In polar coordinates

$$\psi(r, \theta, \phi) = -\frac{m}{2\pi\hbar^2} \frac{\exp(ikr)}{r} \int \exp\left(\frac{i\mathbf{p} \cdot \mathbf{y}}{\hbar}\right) V(\mathbf{y}) d\mathbf{y}$$

§2

Noting that

$$\frac{d\sigma}{d\Omega} = |\psi(r, \theta, \phi)|^2 r^2 = \left| -\frac{m}{2\pi\hbar^2} \int \exp\left(\frac{i\mathbf{p} \cdot \mathbf{x}}{\hbar}\right) V(\mathbf{x}) d\mathbf{x} \right|^2$$

we define $f(\theta, \phi)$ to be the scattering amplitude

$$f(\theta, \phi) = -\frac{m}{2\pi\hbar^2} \int \exp\left(\frac{i\mathbf{p} \cdot \mathbf{x}}{\hbar}\right) V(\mathbf{x}) d\mathbf{x}$$

so that

$$\frac{d\sigma}{d\Omega} = |f(\theta, \phi)|^2$$

For example, for Coulomb potentials we have

$$f(\theta, \phi) = \frac{Z\alpha\hbar c}{2E(1 - \cos\theta)}$$

Hence the Rutherford cross section

$$\frac{d\sigma}{d\Omega} = |f(\theta, \phi)|^2 = \left[\frac{Z\alpha\hbar c}{2E(1 - \cos\theta)} \right]^2$$

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